DESIGN TASK
ELECTRICAL INSTALLATION

GROUP MEMBERS

THARLUS A.P.A (2022/E/004)

ANOJINTH K (2022/E/076)

VITHUSHAN S (2022/E/152)

USMAN M.T(2022/E/095)

YOGABALAN R (2022/E/123)

KARUNANAYAKA H.K.V.S. (2022/E/043)

HERATH H.M.N.P. (2022/E/052)

WIJESINGHE H.M.M.S (2022/E/056)

THILAKARATHNE B.P.T.S. (2021/E/021)

Table of Contents

Aim
Objectives

Introduction

1. Electrical Installation Design

1.1 Introduction
1.2 Objectives
1.3 Design Basis and Assumptions
1.4 Machine Load Calculations
1.5 Total Connected Machine Power
1.6 Total Machine Current
1.7 Socket Outlet Demand Calculation
1.8 Present Total Demand
1.9 Kitchen and Future Requirement
1.10 Design Current and Incomer Cable Selection
1.11 Feeder Cable Voltage-Drop Calculation
1.12 Final-Circuit Voltage-Drop Calculation
1.13 Distribution Board Location and Cable Routing
1.14 Single-Line Diagram and AutoCAD Drawings
1.15 Inspection, Testing and Final Verification
1.16 Discussion and Conclusion

2. Solar PV Installation Design

2.1 Introduction
2.2 Design Summary
2.3 Site Assessment
2.4 PV Panel Selection
2.5 Array Size and Layout
2.6 Structural Considerations
2.7 Inverter Selection
2.8 String Design and Compatibility Checks
2.9 PVsyst Simulation Results
2.10 Monthly Energy and Performance
2.11 Consistency with Earlier Preliminary Design
2.12 Balance of System and Protection
2.13 Cable Sizing and Voltage Drop
2.14 Single-Line Diagram
2.15 Site Verification Before Final Construction
2.16 Assumptions
2.17 Contribution
2.18 Conclusion

3. Lightning Protection System Design

3.1 Introduction
3.2 Applicable Standards
3.3 Building Dimensions
3.4 Selection of Lightning Protection Class
3.5 Selection of Air-Termination System
3.6 Selection of Air-Termination Rod Height
3.7 Protective Angle
3.8 Air-Termination Arrangement
3.9 Protection of Lower Roof
3.10 Down-Conductor System
3.11 Proposed Down-Conductor Layout
3.12 Down-Conductor Selection
3.13 Earth-Termination System
3.14 Type B Ring Earth Electrode
3.15 Earth Electrode Design
3.16 Final Lightning Protection Design
3.17 Drawings
3.18 Conclusion

4. Overall Conclusion

References

Aim

The aim of this design task is to design a safe, reliable, and standards-compliant electrical installation, solar PV system, and lightning protection system for the Mechanical Engineering Workshop, including the required calculations, equipment selection, cable sizing, drawings, and protection arrangements.

Objectives

1. To design a suitable electrical installation system for the Mechanical Engineering Workshop, including demand calculation, cable sizing, voltage-drop calculation, and protective-device selection.
2. To propose a suitable solar PV system by selecting appropriate panels, inverters, electrical components, and grid-connection cables.
3. To design a suitable lightning protection system using the Protective Angle Method for the assigned building.
4. To select appropriate air-termination, down-conductor, and earth-termination systems based on the selected protection class.
5. To prepare the required 2D drawings and single-line diagrams for the proposed electrical systems while ensuring safe and reliable operation.

Introduction

Electrical installation design is an important part of ensuring the safe, reliable, and efficient operation of the Mechanical Engineering Workshop. This design task focuses on developing a suitable electrical installation for the workshop by considering the connected machinery, power demand, three-phase balancing, voltage drop, protective devices, incomer rating, and cable selection. The design also includes the preparation of the required 2D layout and single-line diagram to represent the proposed electrical distribution system.

In addition to the electrical installation, the task includes the design of a solar PV system and lightning protection system for the assigned facility. The solar PV design considers the available roof area, solar potential, panel and inverter selection, layout, electrical protection, and grid connection. The lightning protection design is carried out using the Protective Angle Method, including the selection of the appropriate protection class, air-termination system, down conductors, and earth-termination system. Together, these three design areas provide a complete electrical solution that improves the safety, reliability, and energy performance of the Mechanical Engineering Workshop

1.Electrical Installation

1.1 Introduction

A properly designed electrical installation is essential for the safe and efficient operation of a mechanical workshop. Since different machines require different amounts of electrical power, the distribution system should be designed to provide a reliable supply while minimizing voltage drops, electrical losses, and potential safety hazards.

This section focuses on the electrical installation design of the Mechanical Engineering Workshop at the Faculty of Engineering, University of Jaffna**.** The workshop consists of several electrical machines, including milling machines, lathe machines, a power press, a band saw, drilling machines, and an air compressor, with a total connected machine load of 55.8 kW**.** In addition, 20 socket outlets are considered to support general workshop activities.

The electrical supply is planned to be obtained from an existing panel board located approximately 200 m away from the workshop**.** Therefore, proper cable sizing, voltage drop calculations, and selection of protective devices are important aspects of the design. A suitable distribution board location and cable routing arrangement are also proposed to ensure convenient operation and maintenance.

The design includes load and maximum demand calculations, three-phase load balancing, socket outlet arrangements, cable sizing, voltage drop calculations, and selection of protective equipment. The proposed electrical layout and single-line diagram are also developed using AutoCAD to illustrate the overall distribution system.

The main objective of this section is to propose a safe, practical, and reliable electrical installation that meets the workshop's current power requirements while considering future electrical demand.

1.2 Objectives

- Determine the connected and diversified electrical demand of the workshop.
- Arrange the twenty 13 A socket outlets as six circuits (S1–S6) and distribute them across the three phases.
- Select a suitable workshop distribution board, incomer and upstream panel-board protective device.
- Check voltage drop for the 200 m feeder and for representative final circuits.
- Select practical cable sizes and protective devices for each machine and socket circuit.
- Propose improvements for the kitchen/service area, DB location and machine cable paths.
- Provide a conceptual 2D electrical layout and a single-line diagram suitable for transfer to CAD.

1.3 Design Basis And Assumptions

- Nominal supply: 400/230 V, 50 Hz, three-phase, four-wire.
- Ambient temperature assumed in the submitted work: 30 °C.
- Copper conductors are assumed. Final circuits are treated as multicore thermoplastic cables
- Machine distances Assumptions are taken from the DB.
- Twenty 13 A socket outlets are provided and arranged as six circuits S1-S6.
- The planned socket distribution is: L1 = 4 + 3 = 7 sockets, L2 = 4 + 3 = 7 sockets, and L3 = 3 + 3 = 6 sockets.
- A 50% diversity factor is applied to the socket-outlet connected current, as specified for this assignment calculation.
- A further 15% allowance is added to the calculated present demand to cover kitchen requirements and reasonable future expansion.

1.4 Machine Load Calculations

For three-phase machines, the current is calculated from $I = P/(\sqrt{3}\ V \cos\varphi)$. For the single-phase drilling machine, $I = P/(V \cos\varphi)$.

$$I_b = P/(\sqrt{3}\ V \cos\varphi)$$

$$= 4750/(\sqrt{3} * 400 * 0.9)$$

$$= 7.62\ A$$

Choose C type breaker from the

$$I_n = 10\ A$$

$$I_b < I_n < I_t$$

Using The Table TABLE 4D4A-Multicore annoured 70˚c thermoplastic insulated cables (COPPER CONDUCTORS) we take 1.5 mm^2 cable

All the calculations finished in this way

Table 01: Calculations

Load	Power	pf	Supply	Ib	Protection	Cable	Approx. length
Milling machine	4.75 kW	0.90	3φ	7.62 A	C10 A	1.5 mm²	12 m
Heavy lathe	9.25 kW	0.78	3φ	17.12 A	C20 A	2.5 mm²	10 m
Medium lathe (1-6)	4.35 kW	0.85	3φ	7.39 A	C10 A	1.5 mm²	6-12 m
Power press	6.50 kW	0.88	3φ	10.66 A	D16 A	1.5–2.5 mm²	12 m
Band saw	2.60 kW	0.78	3φ	4.81 A	C6/C10 A	1.5 mm²	10 m
Drilling machine 1,2	1.30 kW	0.73	3φ	2.57 A	C6 A	1.5 mm²	6–8 m
Air compressor	4.00 kW	0.75	3φ	7.70 A	D10 A	1.5 mm²	6 m

1.5 Total Connected Machine Power

$P_{machines} = 4.75 + 9.25 + (6 \times 4.35) + 6.50 + 2.60 + (2 \times 1.30) + 4.00$

$P_{machines} = 4.75 + 9.25 + 26.10 + 6.50 + 2.60 + 2.60 + 4.00$

$P_{machines} = 55.80$ Kw

1.6 Total Machine Current

$I_{machines} = 7.62 + 17.12 + (6 \times 7.39) + 10.66 + 4.81 + (2 \times 2.57) + 7.70$

$I_{machines} = 7.62 + 17.12 + 44.34 + 10.66 + 4.81 + 5.14 + 7.70$

$I_{machines} = 97.39$ A per phase

1.7 20 × 13 A Socket Demand with 50% Diversity

Socket design current on a phase is calculated as:

$I_{socket} = (\text{number of sockets} \times 13\ A) \times 0.50$

L1 current $= (7 \times 13) \times 0.50 = 45.5$ A

L2 current $= (7 \times 13) \times 0.50 = 45.5$ A

L3 current $= (6 \times 13) \times 0.50 = 39.0$ A

Table 02 : Socket Current Calculation

Phase	Planned socket groups	No. sockets	Demand at 50% Diversity
L1	4 + 3	7	45.5 A
L2	4 + 3	7	45.5 A
L3	3 + 3	6	39.0 A
Total		20	

1.8 Present Total Demand Before Kitchen/Future Allowance

L1 = 97.39 + 45.50 = 142.89 A

L2 = 97.39 + 45.50 = 142.89 A

L3 = 97.39 + 39.00 = 136.39 A

Maximum phase demand = 142.89 A

1.9 Kitchen and Future Requirement – 15% Allowance

A 15% allowance is added to the present calculated demand to cover the kitchen requirement and reasonable future load growth. For a conservative incomer selection, the allowance is applied to each phase demand.

Future/kitchen allowance factor = 1 + 15/100 = 1.15

L1 final demand = 142.89 × 1.15 = 164.32 A

L2 final demand = 142.89 × 1.15 = 164.32 A

L3 final demand = 136.39 × 1.15 = 156.85 A

Table 03 : Demand Calculations

Phase	Machine demand	Socket demand	Final demand incl. 15%
L1	97.39 A	45.50 A	164.32 A
L2	97.39 A	45.50 A	164.32 A
L3	97.39 A	39.00 A	156.85 A

1.10 Design Current And Cable Of Incomer

The maximum final phase demand is therefore:

Ib = 164.32 A

For protective-device selection, Ib ≤ In.

From the TABLE 4D4A-Multicore annoured 70^0 C thermoplastic insulated cables (COPPER CONDUCTORS) we can choose 95 mm^2 cable

1.11. Feeder Cable Voltage-Drop Calculation

$$\text{Voltage Drop} = (Ib^* (mV/A/m)L)/1000$$

$$\Delta V \approx (164.32 * 0.135 * 200)/1000 = 4.437 \text{ V}$$

Percentage feeder drop ≈ 4.437 /400 × 100 = 1.11 %.

This leaves useful voltage-drop margin for the final circuits.

1.12. Final-Circuit Voltage-Drop

Table 04 : Voltage Drops

Circuit	Ib	Length	Illustrative mV/A/m	Approx. ΔV
Milling, 1.5 mm²	7.62 A	12 m	25	2.29 V
Heavy lathe, 2.5 mm²	17.12 A	10 m	15	2.57 V
Medium lathe, 1.5 mm²	7.39 A	12 m	25	2.22 V
Power press, 1.5 mm²	10.66 A	12 m	25	3.20 V
Band saw, 1.5 mm²	4.81 A	10 m	25	1.20 V
Drilling, 1.5 mm²,	2.57 A	6 m	25	0.39V
Air compressor, 1.5 mm²	7.70 A	6 m	25	1.16 V

1.13 Distribution board location and cable-routing proposal

1.13.1 Proposed workshop DB position

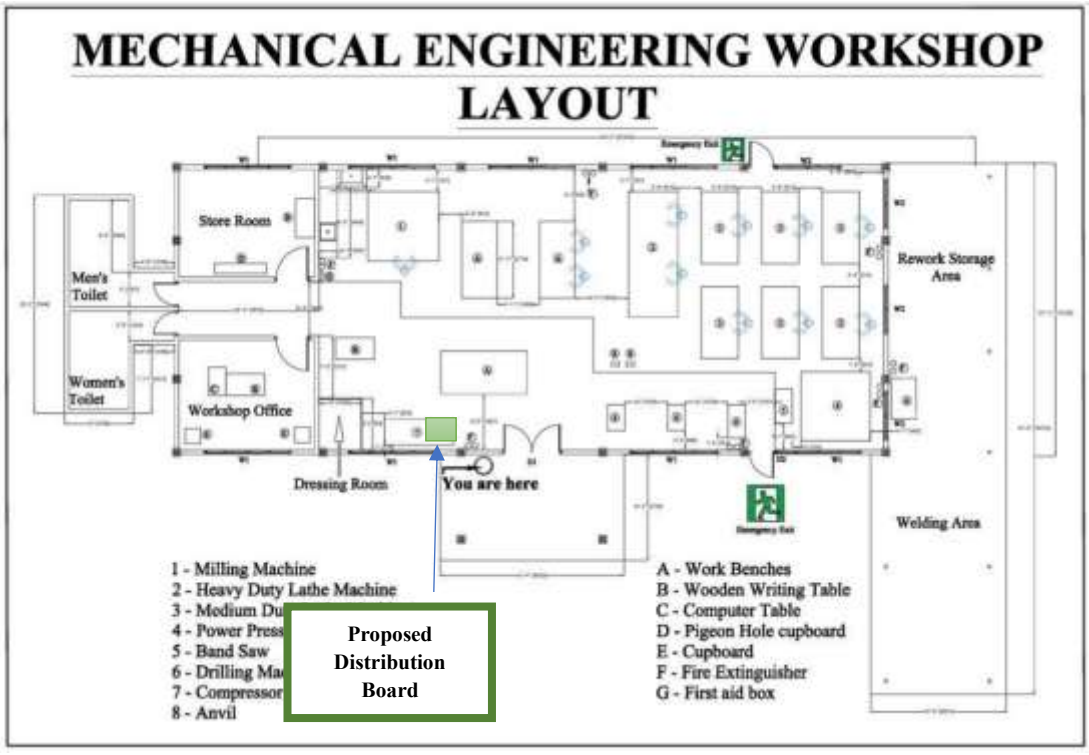


Figure 1.13.1. Proposed workshop distribution board location, as marked by the group on the workshop layout.

The workshop DB is proposed on the internal wall near the main entrance marked “You are here”. The group has identified the position on the workshop drawing. Placing the board near an accessible circulation area should simplify isolation, fault finding and route planning, particularly because both machine and general socket circuits originate at this point.

Before fixing the DB at this location, access clearance, protection against mechanical damage, avoidance of exit obstruction and protection against dust and moisture must be checked on site. The panel should be clearly labeled and positioned so that a person working at it does not block a door or emergency escape route.

1.13.2 Cable route principles

The 200 m incoming feeder should arrive at the workshop through an appropriately protected buried duct or a supported route determined by the site survey. Inside the workshop, machine circuits can follow high-level tray or trunking along the walls and then drop in mechanically protected conduit or flexible industrial cable to individual isolators. The final cable routes should avoid moving-machine envelopes, sharp edges, welding spatter, door swings and locations where workshop materials are routinely stored.

Separate and identifiable circuits are proposed for each fixed machine. Each circuit should terminate at a local lockable isolator located where maintenance personnel can isolate the equipment without crossing the working envelope of another machine. Cable containment and earth bonding must remain continuous throughout.

1.14 Single-line diagram and required AutoCAD figures

The final electrical single-line diagram must be consistent with the circuit schedule. It should identify the source panel, 200 m feeder, workshop main isolator/incomer, the outgoing motor circuits and the six single-phase socket ways. The diagram must also show neutral and PE arrangements, breaking capacities, protective-device settings and selected cable specifications.

The conceptual supply chain is: Existing supply panel → outgoing MCCB → 200 m Cu feeder → Workshop DB → individual machine isolators/motor starters and socket RCBO circuits.

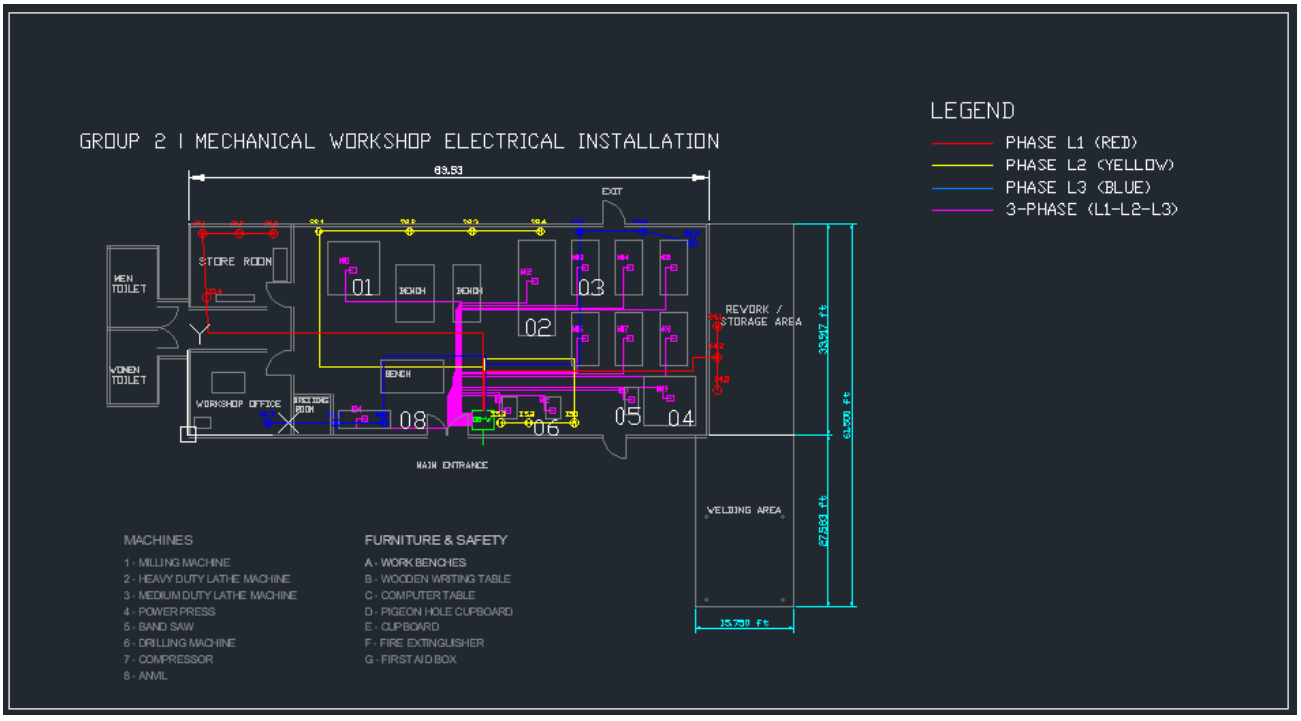


Figure 1.14.1. Complete electrical single-line diagram (SLD)

1.15 Inspection, testing and final verification

After completing the design drawings, the selected devices and installed cable route must be checked against the workshop conditions. Before energizing an installation of this type, competent persons should verify both the physical wiring and the operation of its protective devices. The following list summarizes the remaining evidence required for the report and construction design.

Table 1.15.1: Design completion and commissioning checklist.

Check	Required evidence	Status
Machine nameplates	Supply phase, FLC, efficiency, motor starters	Pending
Route measurements	200 m feeder route and each outgoing cable length	Pending CAD / site survey
95 mm² cable design	Manufacturer Iz and R/X; installation and grouping factors	Pending
Socket radial sizing	Longest route, mV/A/m, Iz and RCBO suitability	Pending
Supply fault level	Source short-circuit current and MCCB Icu/Ics	Pending
Earthing / CPC	Earth-fault path, PE sizing, disconnection time	Pending
Tests	Continuity, insulation, polarity, phase rotation, RCD trip, functional tests	For commissioning
Drawings	Electrical 2D layout and completed SLD	Insert checked exports

1.16 Discussion and conclusion

The electrical installation design for the Mechanical Engineering Workshop was carried out by considering the power requirements of the existing machines, socket outlets, and possible future expansion. The total connected machine load was calculated as 55.8 kW, and the maximum design current was estimated as 164.32 A, including the socket outlet demand and a 15% allowance for future requirements.

Based on the calculations, suitable cable sizes and protective devices were proposed for the machines and socket outlet circuits. A 95 mm² copper cable was selected as the proposed main supply cable for the 200 m distance, with a calculated voltage drop of approximately 1.11% under the stated assumptions. The socket outlets were also distributed across the three phases to maintain a reasonably balanced load.

A suitable distribution board location and cable routing arrangement were proposed by considering safety, accessibility, and ease of maintenance. The proposed electrical layout and single-line diagram help illustrate the overall distribution system and the connections between the main supply, distribution board, and workshop equipment.

.

2. Solar PV Installation Design

2.1 Introduction

This section covers the rooftop solar photovoltaic system for the Mechanical Engineering Department roof (Group 2). The design target is a 200 kW-class grid-connected system. The final array size, inverter selection, string arrangement, balance of system, cabling and expected energy production are based on the completed PVsyst model and the preliminary roof and cable-route information available for the project.

The design is organized in the same sequence as the installation task: site assessment, solar panel selection and layout, inverter and balance-of-system design, cable sizing to the Faculty of Engineering transformer, and energy-yield verification using PVsyst. Assumptions that still require confirmation by site survey or final equipment data are stated clearly.

2.2 Design Summary

Table 2.1: Design summary

Parameter	**Value**
Array	330 x Jinkosolar JKM-615N-66HL4M-BDV modules (615 Wp)
Array capacity	202.95 kWp (approximately 203 kWp)
Module area	Approximately 891 m² total module surface
String layout	22 strings of 15 modules; 11 strings per inverter
Mounting / orientation	Fixed plane, 9.5° tilt and 0° azimuth in the PVsyst variant
Inverters	2 x Sungrow SG110CX-P2, 110 kWac each
DC/AC ratio	202.95 / 220 = 0.923
Grid connection	400 V, three-phase LV connection to the Faculty of Engineering transformer
Main AC feeder	2 parallel runs of 4-core 185 mm² Cu/XLPE/SWA; 100 m preliminary route assumption
Expected energy	324,918 kWh/year; 1,601 kWh/kWp/year; PR = 84.82%

2.3 Site Assessment

The selected site is the roof of the Mechanical Engineering Department. For the completed PVsyst simulation, Terumurikandi, Sri Lanka (9.32° N, 80.41° E, altitude 32 m) is used with Meteonorm 8.1 synthetic weather data. The PV field is modelled as a fixed plane at 9.5° tilt and 0° azimuth. The current PVsyst variant does not include a 3D near-shading scene, so physical shading must still be checked on site.

Table 2.2: Site assessment summary

Item	Assessment / assumption
Roof geometry	Two adjoining preliminary roof sections: 30.56 m x 33.25 m average and 31.40 m x 32.95 m.
Horizontal roof area	Approximately 2,050.75 m² total.
Approx. sloping area	Approximately 2,079.02 m² total.
Roof pitch	atan(100/600) = 9.46°, consistent with the 9.5° PVsyst tilt.
Shading	Current PVsyst model has no 3D near-shading model; trees, parapets, tanks and rooftop equipment require site inspection.
Dust / soiling	Local conditions and a practical cleaning schedule must be confirmed.
Structure	Dead load, wind uplift, mounting points and waterproofing require structural verification.
Access	Walkways, edge clearance and maintenance access must remain available.
Orientation	PVsyst uses 0° azimuth; final north reference and roof orientation must be confirmed on site.

2.4 Panel Selection

The final PVsyst model uses the Jinkosolar JKM-615N-66HL4M-BDV module. It is a 615 Wp N-type monocrystalline bifacial module. The high module power allows the 200 kW-class target to be reached with 330 modules, giving a final installed DC capacity of 202.95 kWp. The completed PVsyst model is treated as the controlling source for the final module type and module count.

Table 2.3: PV module data used for the Group 2 design

Parameter	Final value	Note
Manufacturer	Jinkosolar	Module selected in the completed PVsyst project
Model	JKM-615N-66HL4M-BDV	Final simulated model
Rated power	615 Wp	STC nameplate
Technology	N-type monocrystalline, bifacial	As recorded in the Group 2 design report
Number of modules	330	Final array
Total module surface	Approximately 891 m²	PVsyst array surface
Installed DC capacity	202.95 kWp	330 x 615 Wp

2.5 Array Size and Layout

The 200 kW target cannot be met exactly with an integer number of 615 Wp modules. Using 330 modules gives 202.95 kWp, which is 2.95 kWp (1.48%) above 200 kWp. Therefore the system is reported as a 200 kW-class design. The electrical array is divided into two identical sub-arrays so that each inverter receives 165 modules.

Table 2.4: Array and string configuration

Sub-array	Modules	String arrangement	DC capacity	Inverter
PV Array #1	165	11 strings x 15 modules	101.475 kWp	1 x SG110CX-P2
PV Array #2	165	11 strings x 15 modules	101.475 kWp	1 x SG110CX-P2
Total	330	22 strings	202.95 kWp	2 x SG110CX-P2



Figure 2.5.1: Final AutoCAD Rooftop PV Module and String Layout – 330 Modules

2.6 Structural Considerations

The roof layout is still preliminary. Before construction, the roof structure should be checked for the PV module and mounting-system dead load, wind uplift, fixing points and waterproofing requirements. The final CAD layout should preserve access paths and edge clearances and should avoid obstructions such as tanks, vents, parapets and other rooftop equipment.

2.7 Inverter Selection

The final design uses two Sungrow SG110CX-P2 three-phase string inverters, each rated at 110 kWac. This is the inverter model used in the completed PVsyst simulation. The two-inverter arrangement matches the two equal 101.475 kWp sub-arrays and provides a total AC rating of 220 kW.

Table 2.5: Inverter data used in the final design

Inverter parameter	Value	Source / design use
Model	Sungrow SG110CX-P2	Completed PVsyst configuration
Quantity	2	One inverter per 165-module sub-array
Rated AC power	110 kW each	220 kW total
PV operating range	180-1000 V DC	PVsyst / Group 2 report
Maximum PV input voltage	1100 V DC	Group 2 report
MPPT inputs	12 per inverter	Group 2 report
Grid side	400 V, three-phase	Group 2 report
DC/AC ratio	0.923	202.95 kWp / 220 kWac

2.8 String Design and Compatibility Checks

Each electrical string contains 15 modules in series. In PVsyst, each 165-module sub-array is represented as 11 strings of 15 modules. At the reported 50°C operating condition, each sub-array operates at approximately 568 V and 166 A, with about 94.1 kWp available at MPP. The 568 V operating voltage lies within the inverter operating range of 180 to 1000 V DC.

Table 2.6: String and inverter compatibility checks based on the supplied design data

Check	Calculated / simulated value	Limit / requirement	Result
Modules per string	15	Final PVsyst string configuration	Accepted in simulation
Sub-array MPP voltage at 50°C	568 V	180-1000 V operating range	Pass
Sub-array MPP power at 50°C	94.1 kWp	110 kWac inverter rating	Compatible
DC/AC ratio	0.923	Final selected inverter configuration	No PVsyst clipping loss reported
Cold-string Voc / input-current verification	Not stated in the supplied reports	Final manufacturer datasheets required	To be checked before construction

The PVsyst loss diagram reports 0.00% loss from maximum inverter input current, 0.00% loss over nominal inverter voltage and 0.00% loss over nominal inverter power. However, the supplied project documents do not state all minimum-temperature string-voltage and inverter input-current values required for a complete construction-stage datasheet check; these should be verified using the final manufacturer datasheets.

2.9 PVsyst Simulation Results

The final system was simulated in PVsyst V7.4.7 using Meteonorm 8.1 synthetic weather data for Terumurikandi. The model uses a fixed plane at 9.5° tilt and 0° azimuth, the Perez transposition model, a free horizon and no near-shading model. The annual output predicted by this completed variant is 324,918 kWh injected into the grid.

Table 2.7: PVsyst simulation summary

PVsyst result	**Value**
PV system power	203 kWp
Array	330 x JKM-615N-66HL4M-BDV
Inverters	2 x SG110CX-P2
Total AC power	220 kWac
Global horizontal irradiation	1,868.7 kWh/m²/year
Global incident irradiation	1,887.4 kWh/m²/year
Produced energy	324,918 kWh/year
Specific production	1,601 kWh/kWp/year
Performance ratio	84.82%

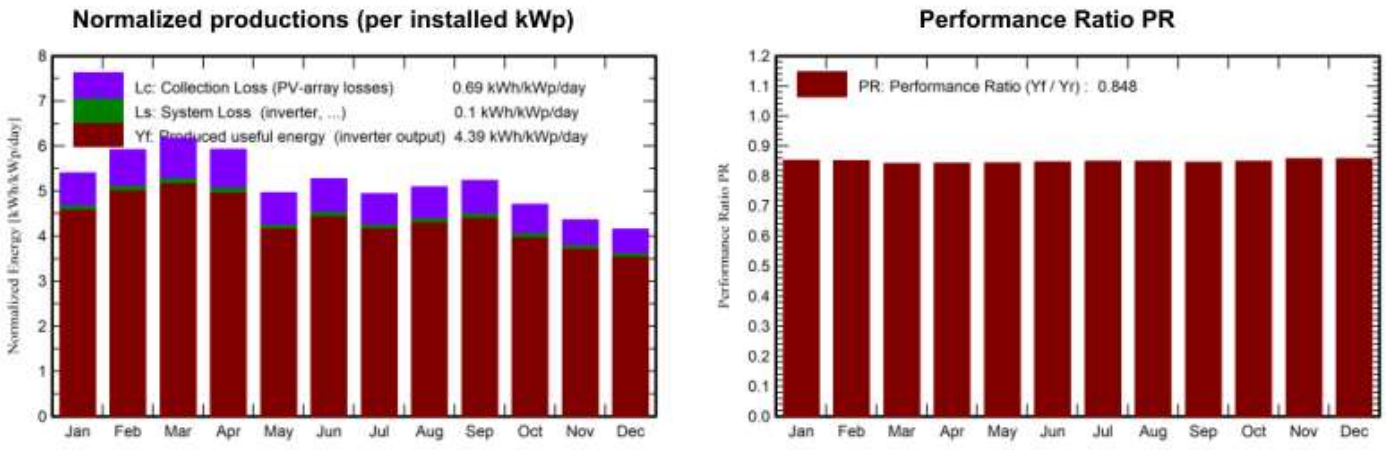


Figure 2.9.1: Normalised production and performance ratio by month (PVsyst)

2.10 Monthly Energy and Performance

Table 2.8: PVsyst monthly results (GlobHor: global horizontal irradiation; GlobInc: global incident irradiation in the array plane; E_Grid: energy injected into the grid)

Month	GlobHor (kWh/m²)	T_Amb (°C)	GlobInc (kWh/m²)	E_Grid (kWh)	PR
January	154.7	26.31	167.5	28,956	0.852
February	156.6	27.18	165.6	28,602	0.851
March	187.8	29.03	191.4	32,675	0.841
April	180.1	30.01	177.7	30,364	0.842
May	159.9	30.33	153.7	26,330	0.844
June	166.8	29.64	158.1	27,165	0.847
July	159.8	29.89	153.1	26,373	0.849
August	161.4	29.23	157.9	27,215	0.849
September	156.5	28.71	157.2	26,960	0.845
October	141.3	28.48	145.9	25,128	0.849
November	123.8	26.98	130.7	22,741	0.857
December	120.1	26.56	128.7	22,409	0.858
Year	1,868.7	28.54	1,887.4	324,918	0.848

The monthly performance ratio remains relatively stable, from about 0.841 to 0.858. Annual grid energy is highest in March at 32,675 kWh and lowest in December at 22,409 kWh, following the seasonal change in incident solar irradiation.

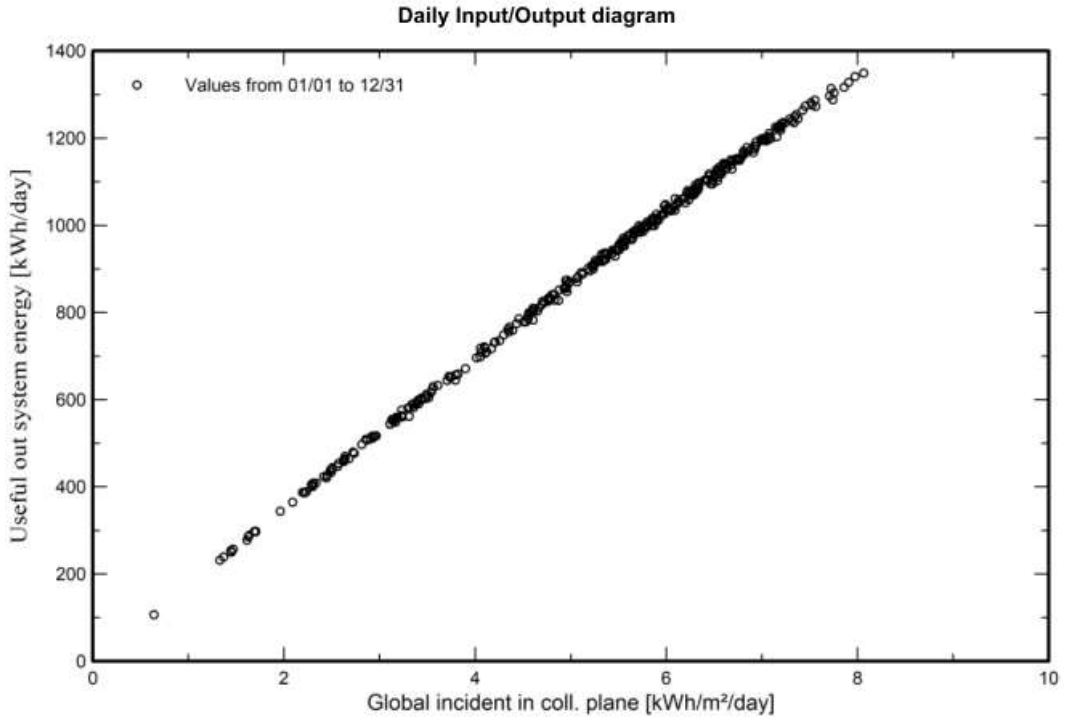


Figure 2.10.1: Daily input/output diagram - useful system energy against incident irradiation (PVsyst)

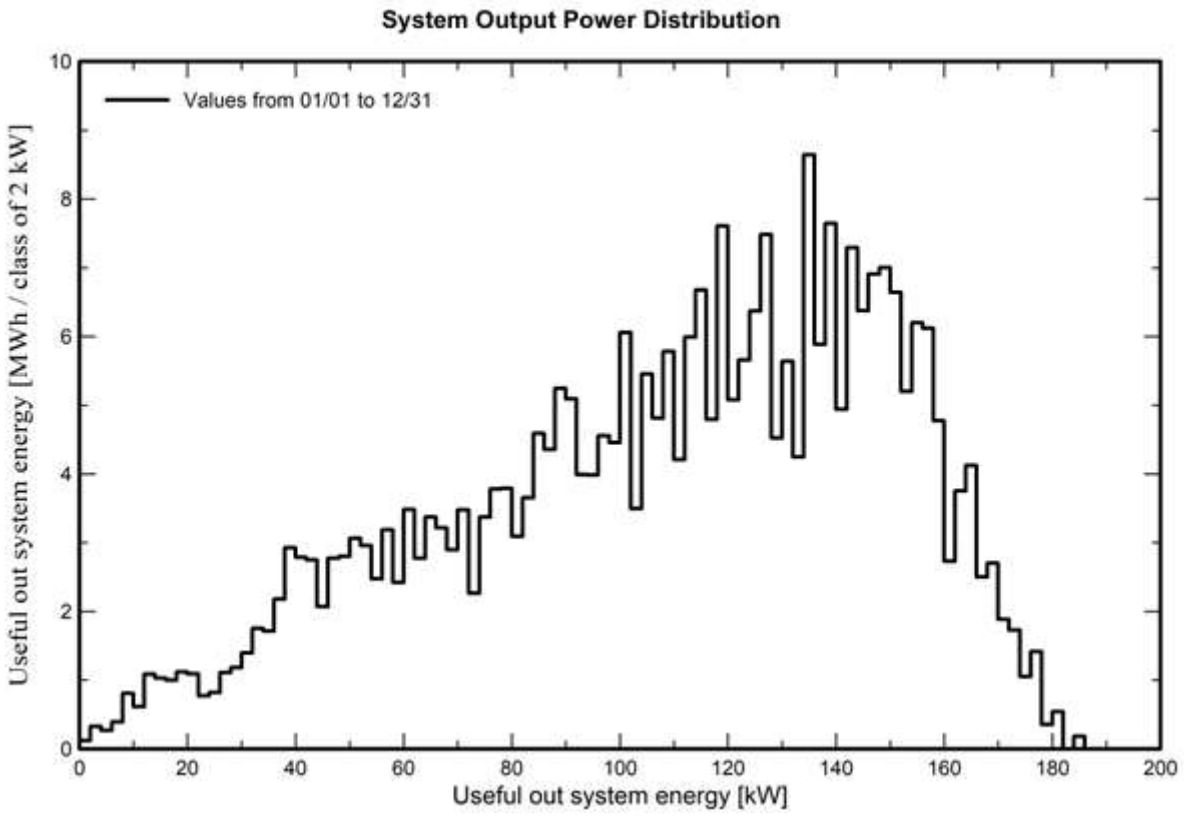


Figure 2.10.2: System output power distribution (PVsyst)

The daily input/output relationship is nearly linear over the main operating range, showing that daily useful energy increases consistently with irradiation. The power-distribution plot shows most annual energy occurring over a broad mid-to-high inverter output range, with very little operation near the 220 kW combined inverter rating.

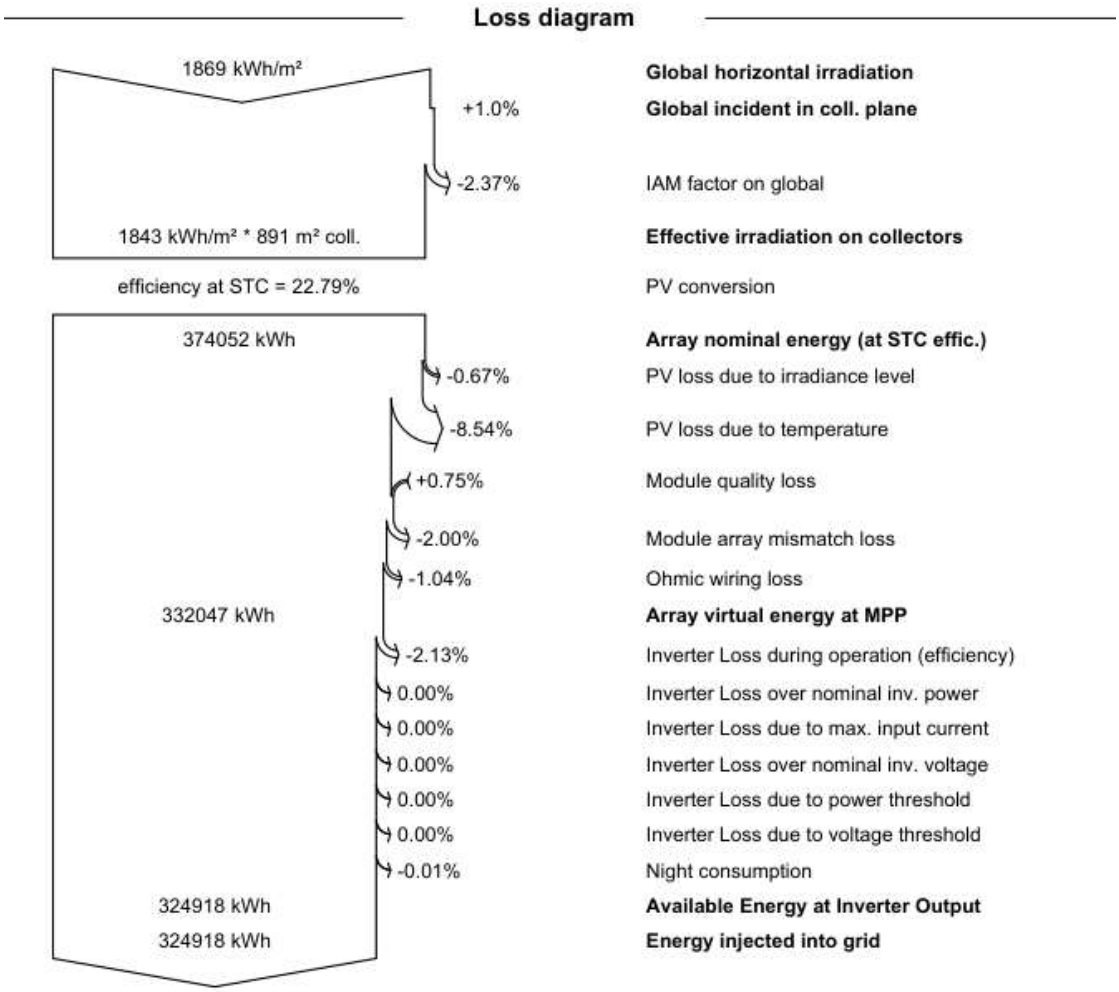


Figure 2.10.3: PVsyst loss diagram for the Group 2 system

The largest reported system loss is the PV temperature loss at 8.54%. Other important losses are the IAM loss at 2.37%, module mismatch at 2.00%, ohmic wiring loss at 1.04% and inverter operating loss at 2.13%. The array virtual energy at MPP is 332,047 kWh, and 324,918 kWh is available at the inverter output and injected into the grid.

2.11 Consistency with the Earlier Preliminary Design

Table 2.9: Preliminary design compared with the final PVsyst-based configuration

Earlier preliminary value	Corrected final value
Jinko 625 Wp module	Jinkosolar JKM-615N-66HL4M-BDV, 615 Wp
320 modules / 200.00 kWp	330 modules / 202.95 kWp
2 x SG100CX-P2	2 x SG110CX-P2
Approximately 199.98 kW AC	220 kW AC
16 modules/string, 10 strings/inverter	15 modules/string, 11 strings/inverter
Approximately 314.4 MWh/year reference estimate	324.918 MWh/year completed PVsyst result
Old cable basis tied to 200 kW configuration	Updated to the 2 x 110 kW inverter configuration

The final report therefore follows the completed PVsyst configuration rather than the earlier 625 Wp / SG100 preliminary calculation sheet. The final module count, inverter type, string arrangement and annual energy result should be kept consistent across the written report, CAD layout and single-line diagram.

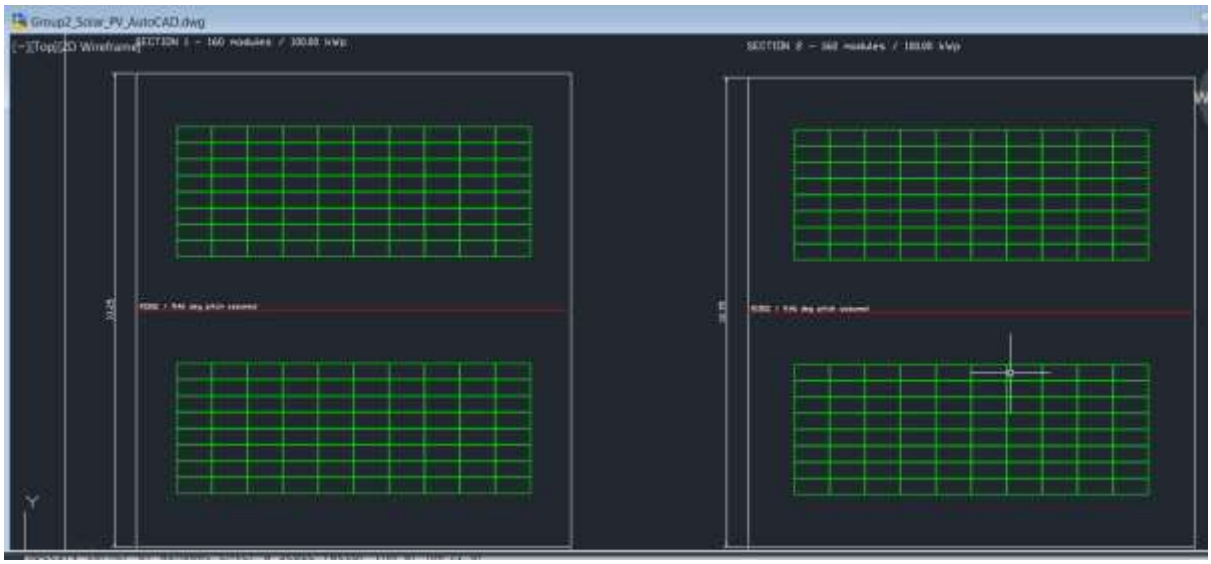


Figure 2.11.1: Preliminary Rooftop PV Module Layout – 320 Modules

2.12 Balance of System and Protection

Table 2.10: Balance of system and protection requirements

Component	Selection / design requirement
PV DC cable	UV-resistant PV cable sized from string current, installation method and derating factors.
PV connectors	Compatible DC connectors with suitable voltage/current rating and correct crimping.
String protection	Final string fuse requirement to be checked from module reverse-current and inverter input data.
DC isolation	DC isolation at the inverter/array interface as required.
DC surge protection	Type II DC SPD; Type I+II to be considered together with the lightning-risk assessment.
AC protection	MCCB for each inverter feeder; final rating and breaking capacity subject to fault-level and coordination checks.
Main AC protection	Main feeder protection based on combined inverter output current and the final installation conditions.
Earthing and bonding	Module frames, mounting structure and inverter protective earth bonded to the building earthing system.
Monitoring / metering	Inverter monitoring plus approved bidirectional export meter at the grid-coupling point.

The protection ratings shown above remain design-stage requirements because the supplied project documents do not include the final transformer fault level, installation method, final cable route or complete manufacturer protection data. These items must be coordinated before construction approval.

2.13 Cable Sizing and Voltage Drop to the Grid Coupling Point

The generated AC power is to be transmitted from the inverter system to the Faculty of Engineering transformer. Because the actual route length has not yet been measured, the design uses a 100 m one-way route as a preliminary assumption. For one 110 kW inverter at 400 V three phase, the rated output current used in the report is approximately 158.8 A. For two inverters, the combined current is approximately 317.6 A.

Table 2.11: Preliminary main AC feeder design

Item	Preliminary design value
Inverters	2 x 110 kW
Voltage	400 V, three-phase
Combined current	Approximately 317.6 A
Assumed route	100 m one way
Main feeder	2 parallel runs of 4-core 185 mm² Cu/XLPE/SWA

Item	Preliminary design value
Current per parallel run	Approximately 158.8 A
Estimated voltage drop	Approximately 3.45 V
Voltage drop percentage	Approximately 0.86% at 400 V

The calculation basis used in the Group 2 report is: $\Delta V = \sqrt{3} I L (R \cos\varphi + X \sin\varphi)$.

The preliminary cable data are R90 approximately 0.129 ohm/km and X approximately 0.0720 ohm/km, with power factor 0.99. The 185 mm² parallel arrangement is retained as a conservative preliminary proposal. Final ampacity, voltage drop and short-circuit withstand must be checked for the actual route, installation method, grouping, ambient temperature and transformer termination.

2.14 Single Line Diagram

The PVsyst single-line diagram represents the final electrical arrangement. Each 165-module sub-array consists of 11 strings of 15 JKM-615N-66HL4M-BDV modules and feeds one 110 kVA SG110CX-P2 inverter. The outputs of the two inverters are combined at the grid injection point. The construction drawing should add the final DC/AC isolation, protection, metering, cable sizes, earthing and connection-point details.

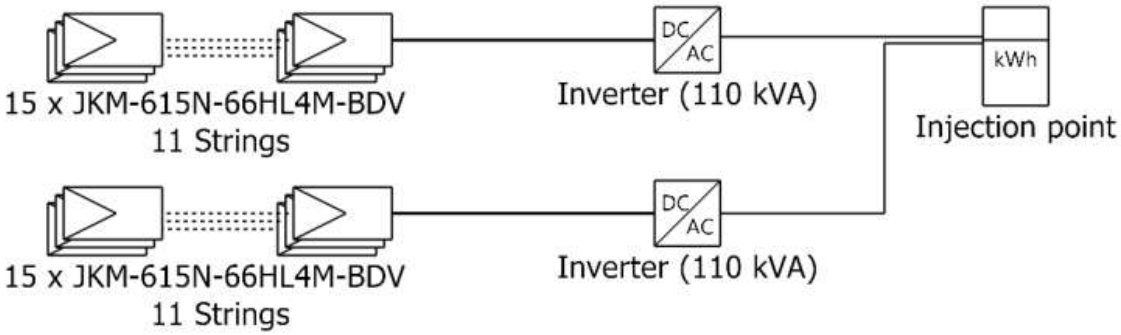


Figure 2.14.1: PVsyst single-line diagram for the final Group 2 simulation variant



Figure 8: AutoCAD Single-Line Diagram of the Final Group 2 Solar PV System

2.15 Site Verification Before Final Construction

- Confirm roof dimensions and ridge direction using the latest CAD drawing or a physical site survey.
- Inspect shading from trees, adjacent buildings, parapets, tanks and rooftop equipment.
- Confirm local dust/soiling conditions and establish a cleaning and maintenance schedule.
- Confirm roof material, structural capacity, wind uplift resistance, mounting method and waterproofing details.
- Measure the actual cable route from the inverter/AC panel to the Faculty of Engineering transformer.
- Obtain transformer spare capacity, fault level, protection settings and utility interconnection requirements.
- Finalize cable installation method and all derating factors.
- Finalize module layout, walkways, edge clearances and maintenance access.

2.16 Assumptions

Table 2.12: Design assumptions requiring final confirmation

Assumption	**Value / basis**	**To be confirmed by**
Roof geometry	Preliminary dimensions from maps / supplied geometry; about 2,050.75 m² horizontal area	Final CAD drawing or site survey
Mounting / tilt	Fixed plane at 9.5° in PVsyst	Final roof geometry and mounting design
Azimuth	0° used in PVsyst	Site orientation / north reference
Shading	No 3D near-shading model in current variant	Physical shading survey and 3D model if required
Weather data	Meteonorm 8.1 synthetic data for Terumurikandi	Optional comparison with measured / other solar-resource datasets
Distance to transformer	100 m one-way	Site route measurement
Feeder installation	2 parallel 4-core 185 mm² Cu/XLPE/SWA preliminary proposal	Actual installation method and derating factors
Transformer / utility data	Not included in the supplied design data	Faculty transformer data and utility requirements
Equipment compatibility	Based on final PVsyst model and stated report parameters	Final manufacturer datasheets before construction

2.17 Contribution

Table 2.13: Contribution to the Solar PV section

Solar PV work item	**Contributor**
Site assessment, roof measurements and solar-resource review	2022/E/076 , 2022/E/152
PV module selection and array sizing	2022/E/095
PV layout and string configuration	2022/E/095 , 2022/E/076
Inverter selection and MPPT/string compatibility	2022/E/095 , 2022/E/152
Balance of System and electrical protection proposal	2022/E/095 , 2022/E/152
Cable sizing and voltage-drop calculation	2022/E/076
PVsyst system setup and simulation	2022/E/095
Solar PV technical documentation and calculations	2022/E/152

Conclusion

A 200 kW-class grid-connected rooftop PV system has been developed for the Mechanical Engineering Department roof. The completed PVsyst configuration uses 330 Jinkosolar 615 Wp bifacial modules, giving 202.95 kWp DC, connected through two Sungrow SG110CX-P2 110 kW string inverters. The array is arranged as two 165-module sub-arrays, each with 11 strings of 15 modules.

The PVsyst model predicts 324,918 kWh/year of grid-injected energy, a specific production of 1,601 kWh/kWp/year and a performance ratio of 84.82%. The largest simulated loss is the 8.54% temperature loss. A preliminary grid feeder of two parallel 4-core 185 mm² Cu/XLPE/SWA runs over an assumed 100 m route gives an estimated voltage drop of approximately 0.86%.

The design is suitable as a developed preliminary Solar PV contribution, but construction-stage approval still requires final roof and shading surveys, structural verification, exact azimuth confirmation, complete manufacturer compatibility checks, actual cable-route measurement, transformer-capacity and fault-level verification, and utility interconnection approval.

Sources

- EC9180 – Electrical Installation Design task requirements for Group 2 Solar PV installation on the Mechanical Engineering Department roof.
- PVsyst V7.4.7 simulation report: EC9180_Group2_Mechanical_Workshop_200kW, Variant VC0, simulation date 10 June 2026.
- Jinkosolar JKM-615N-66HL4M-BDV module data as selected in the completed PVsyst project and recorded in the supplied Group 2 report.
- Sungrow SG110CX-P2 inverter data as selected in the completed PVsyst project and recorded in the supplied Group 2 report.
- Group 2 preliminary Solar PV calculation information as corrected in the supplied final report.

Design Status

Table 14: Current design status

Item	**Status**
Solar PV sizing	Completed preliminary design
PVsyst simulation	Completed for the current VC0 configuration
Roof measurements	Preliminary; final CAD/site survey required
Shading	Not included in current PVsyst; physical inspection / model required
Structural verification	Pending
Cable route	100 m assumed; site measurement required
Transformer / utility approval	Pending
Lightning protection	Excluded from this Solar PV contribution

2. Lightning Protection System Design

2.1 Introduction

The lightning protection system (LPS) is designed to protect the Mechanical Engineering Workshop against the physical effects of a direct lightning strike. An external LPS consists of three main parts: air-termination system, down-conductor system and earth-termination system.

The proposed system is designed using the Protective Angle Method, as specified for Group 2 in the assignment. The air-termination system intercepts the lightning discharge, the down conductors provide a controlled path for the lightning current, and the earth-termination system disperses the current into the ground.

2.2 Applicable Standard

The lightning protection design is based on the following standards:

- SLS 1472 – Protection Against Lightning
- IEC 62305 – Protection Against Lightning

The course lecture identifies SLS 1472 and IEC 62305 as the relevant lightning-protection standards. IEC 62305-3 covers protection against physical damage to structures and life hazards and provides requirements for the design and installation of an LPS.

2.3 Building Dimensions

The dimensions are obtained from the provided AutoCAD drawing. The assignment drawing specifies the dimensions in **centimetres**.

Parameter	Dimension	Converted value
Building length	8000 cm	80 m
Main building width	1200 cm	12 m
Main roof/eave level	1830 cm	18.3 m
Main roof ridge level	1930 cm	19.3 m
Lower roof level	890 cm	8.9 m
Lower level	840 cm	8.4 m
Horizontal extension	850 cm	8.5 m

The 80 m × 12 m main building geometry is therefore used for positioning the air-termination system.

2.4 Selection of Lightning Protection Class

Selected LPS Class: Class III

A detailed lightning-risk assessment is not provided as part of the assignment. Therefore, LPS Class III is selected as an engineering assumption for this academic design.

The selection should be verified by a formal lightning-risk assessment for an actual installation.

For LPS Class III, the IEC 62305 design parameters include:

- Rolling-sphere radius = 45 m
- Mesh size = 15 m × 15 m
- Typical preferred down-conductor spacing = 15 m

The rolling-sphere radius is not used for the present calculation, because the assignment specifically requires the Protective Angle Method. The 45 m value is therefore included only as a Class III reference parameter.

2.5 Selection of Air-Termination System

The air-termination system is selected according to the Protective Angle Method.

The air-termination system consists of vertical air-termination rods installed at suitable positions on the roof. The rods are interconnected with the lightning protection conductor system and connected to the down conductors.

The air terminals are positioned at exposed/high points of the building so that the complete roof and other exposed portions of the structure fall within the required protection volume.

The Protective Angle Method determines the protected zone using the angle formed between the vertical axis of the air terminal and the boundary of the protection zone.

2.6 Selection of Air-Termination Rod Height

For the design, an air-terminal height of h=2m above the relevant roof reference plane is assumed.

This is an engineering assumption for our academic design.

An important point is that h is measured from the air-terminal tip to the reference plane being protected. Therefore, the protective angle must be reconsidered when protecting a lower roof or another lower-level surface.

2.7 Protective Angle

For the selected LPS Class III and the assumed air-terminal height of 2 m, the protective angle used for this design is:

$\alpha = 74$

Protection radius at roof level

For the roof surface at the reference plane, the horizontal protection distance can be determined from:

$R=h \tan \alpha$

where:

$h=2$

and

$\alpha = 74$

Therefore,

$R=h \tan \alpha = 6.97m$

Thus, a 2 m high air-termination rod provides approximately 6.97 m horizontal protection distance at the relevant roof reference plane.

2.8 Air-Termination Arrangement for the 80 m Roof

The main roof dimensions are:

$L=80m$, $W=12m$

The air terminals are distributed along the roof because a single air rod at the centre of the 80 m building would not provide sufficient longitudinal coverage.

The roof width is:

12m

Half of the roof width is:

$12/2=6m$

The calculated protection distance is:

$r=6.97m$

Since:

$6.97>6$

the protection zone extends beyond both sides of the 12 m roof width when the air terminal is positioned along the centre line.

Therefore, the critical direction for the air-terminal arrangement is the 80 m longitudinal direction.

The maximum spacing between adjacent air terminals based on overlapping protection zones is approximately:

Smax=2r

Smax=2(6.97)

Smax=13.94m

Therefore, the air terminals should be distributed along the roof with a spacing not exceeding approximately 13.94 m for this simplified roof-level protective-angle calculation.

Proposed arrangement

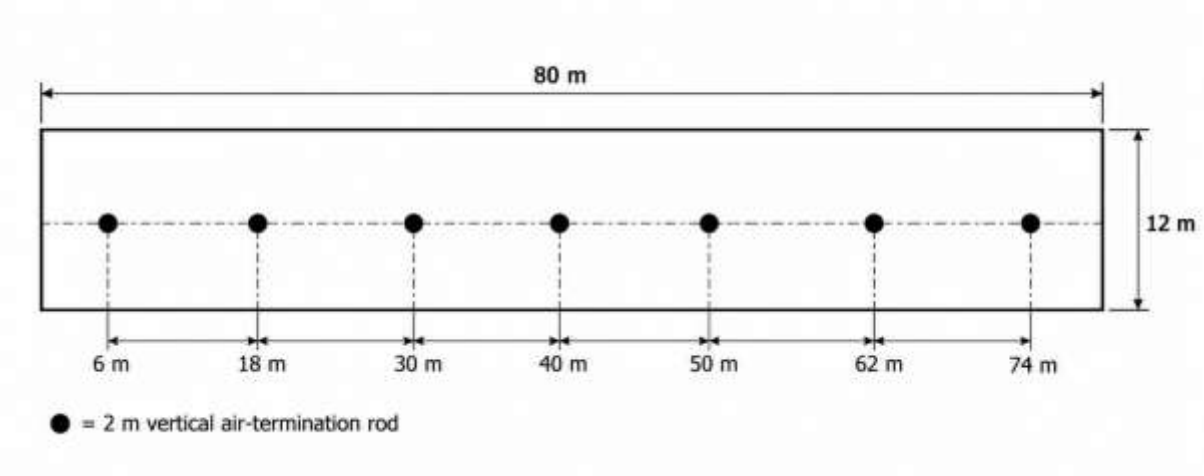
A practical arrangement is to use seven vertical air terminals along the roof centre line at approximately:

$$6\,m,\quad 18\,m,\quad 30\,m,\quad 40\,m,\quad 50\,m,\quad 62\,m,\quad 74\,m$$

from one end of the building.

This arrangement keeps the air-terminal spacing within the calculated protection distance and provides coverage toward both ends of the building.

Conceptual arrangement



The exact locations can be adjusted in the AutoCAD drawing according to roof geometry, structural members and other exposed objects, while maintaining the required protection envelope.

2.9 Protection of the Lower Roof

The lower roof cannot simply be considered protected using the 2 m roof-level calculation.

The protective angle depends on the height of the air terminal above the particular reference plane. Therefore, the lower roof must be separately checked using the protective-angle method.

For example, considering the main roof ridge at:

19.3m

and the lower roof at:

8.9m

the vertical difference is:

19.3-8.9=10.4m

With the 2 m air rod:

h_{lower} =10.4+2

= 12.4m

The corresponding protective angle for this greater value of \(h\) must be obtained from the applicable protective-angle relationship.

The side-elevation AutoCAD drawing should therefore show the protective envelope extending from the air terminals and confirm that the lower roof and other exposed portions remain within the protected volume.

If any portion of the lower roof lies outside the protection envelope, an additional air terminal shall be provided.

2.10 Down-Conductor System

The down-conductor system provides a controlled path for the lightning current from the air-termination system to the earth-termination system.

The down conductors shall:

- Be connected to the air-termination system.
- Be connected to the earth-termination system.
- Follow the shortest practical route.
- Be as straight and vertical as practicable.
- Be distributed around the building perimeter.
- Be located preferably at exposed corners and suitable points around the structure.

For a non-isolated LPS, at least two down conductors are required.

For LPS Class III, the typical preferred spacing is approximately 10m between down conductors.

The 10m value is a typical preferred spacing, rather than a requirement to place conductors at exactly 10 m in every location.

2.11 Proposed Down-Conductor Layout

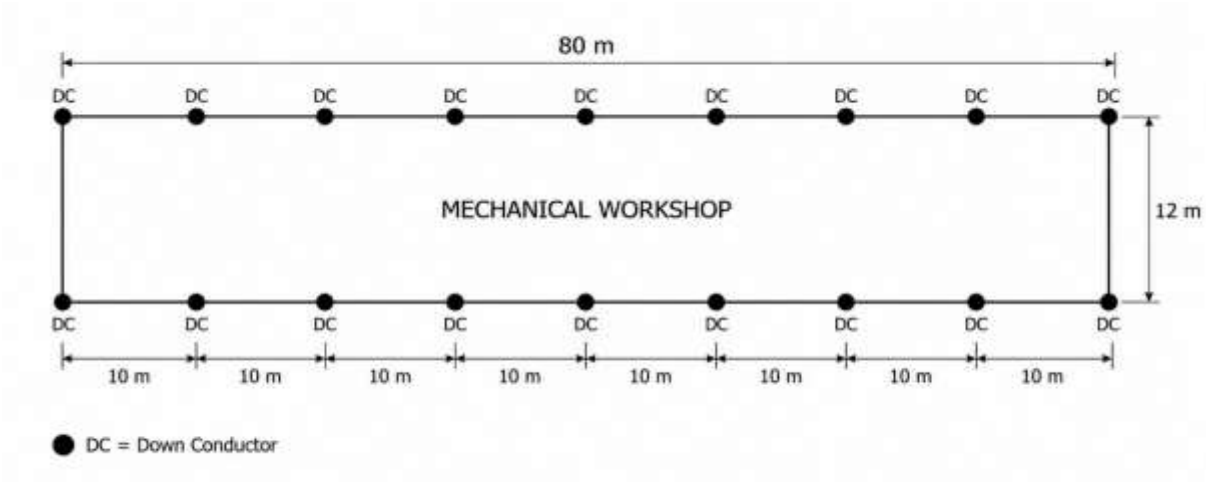
Since the building is approximately 80 m long, two down conductors at the two ends would not provide an appropriate distributed arrangement.

The proposed down-conductor system should therefore be distributed around the building perimeter.

The arrangement should include:

1. Down conductors at the major building corners.
2. Additional down conductors along the 80 m sides.
3. Down conductors associated with significant changes in building geometry.
4. Connections from the roof air-termination system to the down conductors.
5. Direct connections from the down conductors to the Type B earth ring.

A conceptual arrangement is:



DC = Down Conductor

The final number and exact positions should be shown on the roof plan/elevation based on the actual building perimeter.

2.12 Down-Conductor Selection

Copper down conductors are selected, with the final conductor cross-section and component specification selected in accordance with the applicable SLS 1472/IEC 62305 requirements and the relevant lightning-protection component standard**.**

2.13 Earth-Termination System

The earth-termination system provides the path for the lightning current to dissipate into the surrounding soil.

The lecture identifies the principal earth-termination arrangements as:

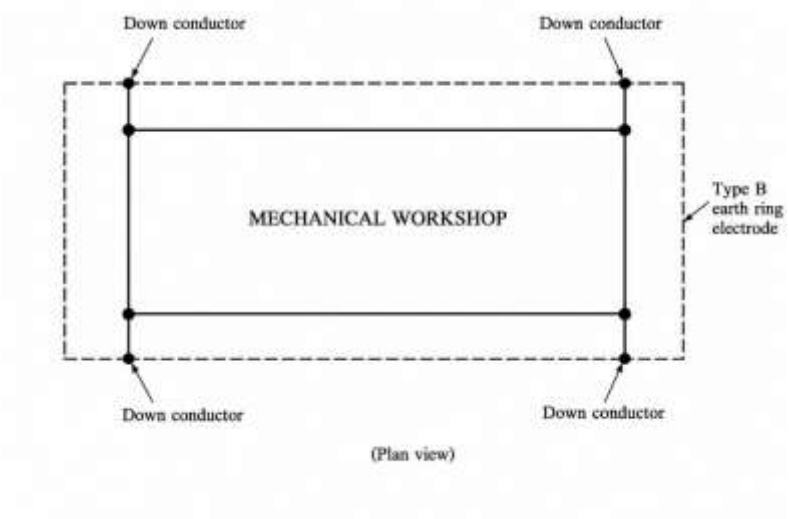
- **Type A**
- **Type B**
- Foundation earth electrode.

For this building, a Type B arrangement is selected.

This is appropriate because the proposed LPS contains multiple down conductors. A Type B arrangement consists of a ring electrode external to the structure or a foundation earth electrode.

2.14 Type B Ring Earth Electrode

The proposed Type B system consists of a **closed ring earth electrode surrounding the building**.The ring electrode is connected to all down conductors.



The Type B ring electrode should be installed around the external perimeter of the building and connected to the down-conductor terminations.

2.15 Earth Electrode Design

A Type B earth-termination system is selected for the proposed lightning protection system. The system consists of a closed ring earth electrode installed around the building and connected to all down conductors. The final dimensions of the earth electrode depend on the soil resistivity, selected LPS protection class, geometry of the ring electrode, required electrode length, and the arrangement of any additional electrodes. Therefore, the exact electrode dimensions and additional vertical or horizontal electrodes will be determined based on the measured site soil resistivity and verified through earth-termination testing after installation.

.

2.16 Drawings of the system

A. Front Elevation



Figure 2.16.1: Before the editing front view

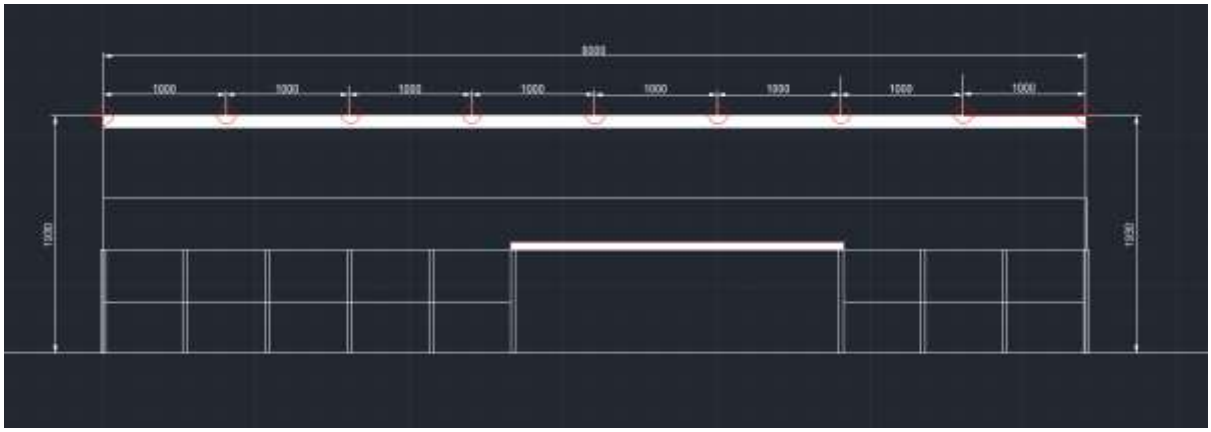


Figure 2.16.2 : Front view with dimension

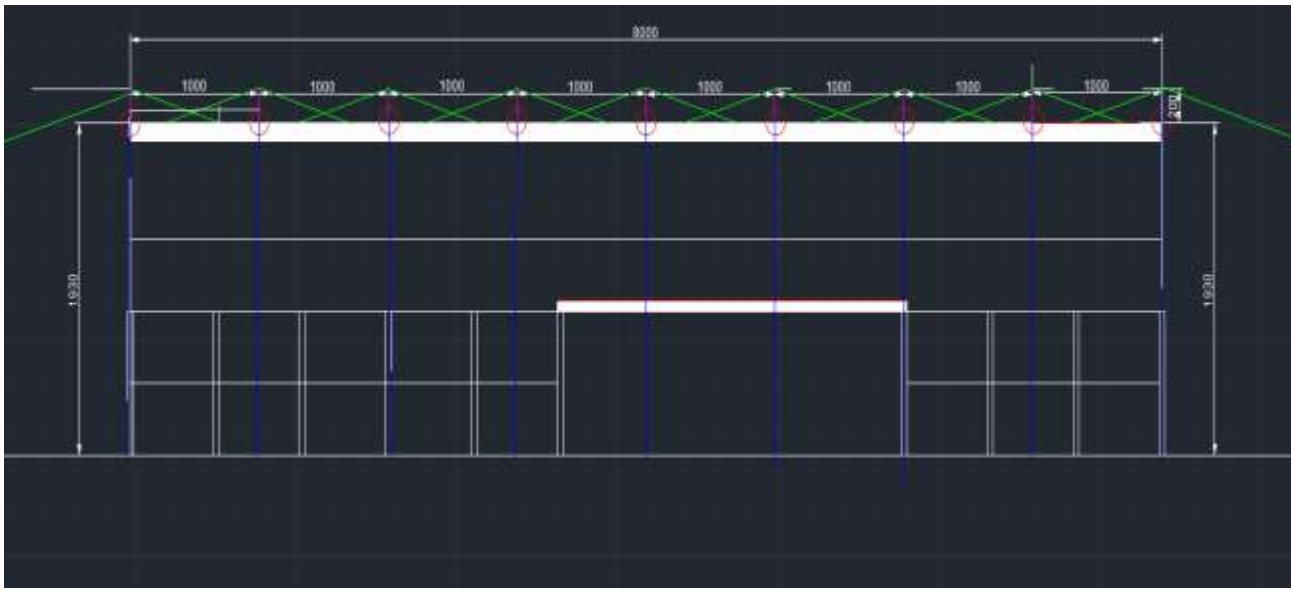


Figure 2.16.3: Protection rod and wiring included

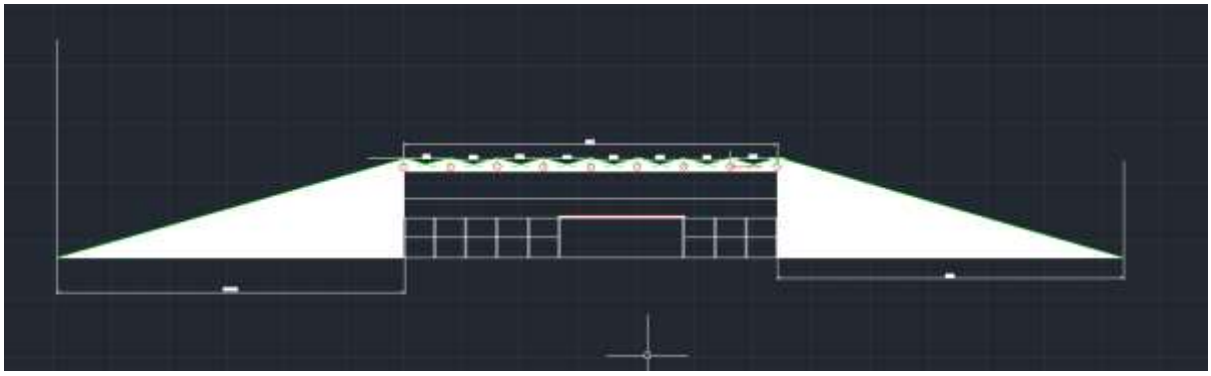


Figure 2.16.4: Protection area included for front view

B. Side Elevation

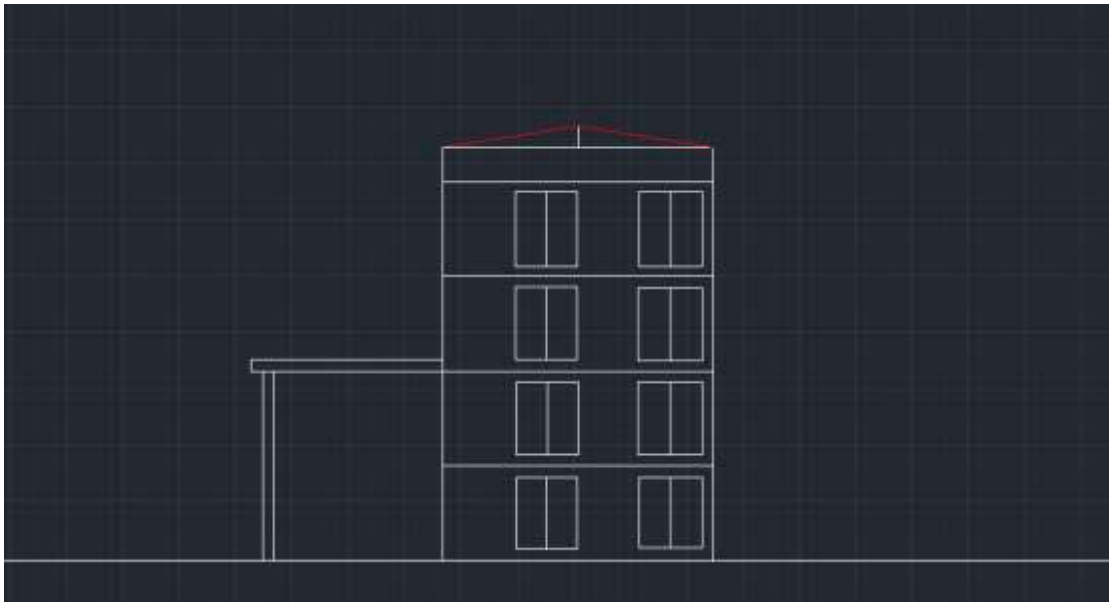


Figure 2.16.5: Raw side view before editing

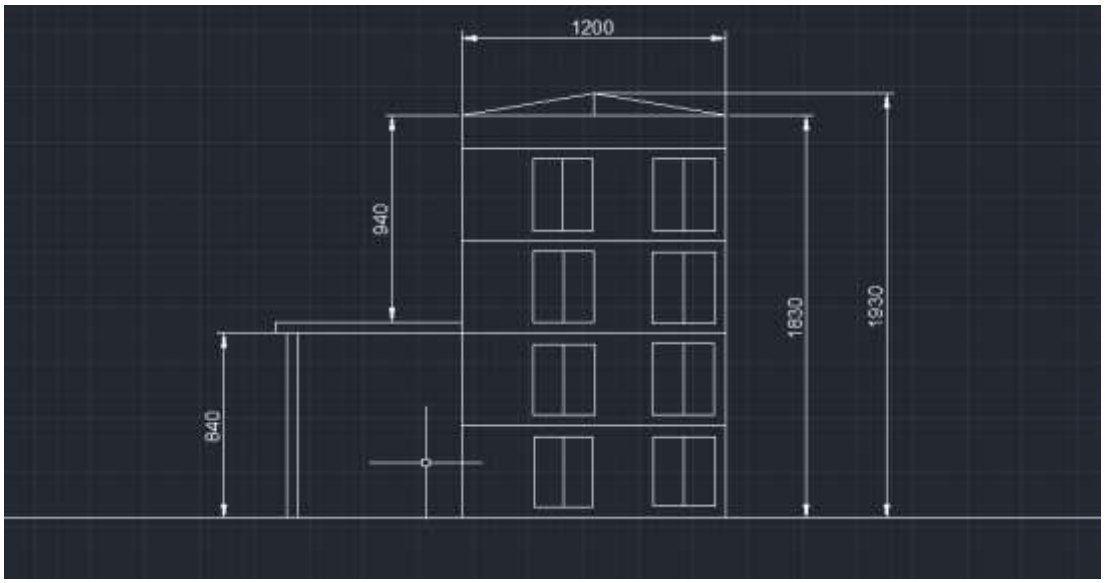


Figure 2.16.5: Side view with dimensions

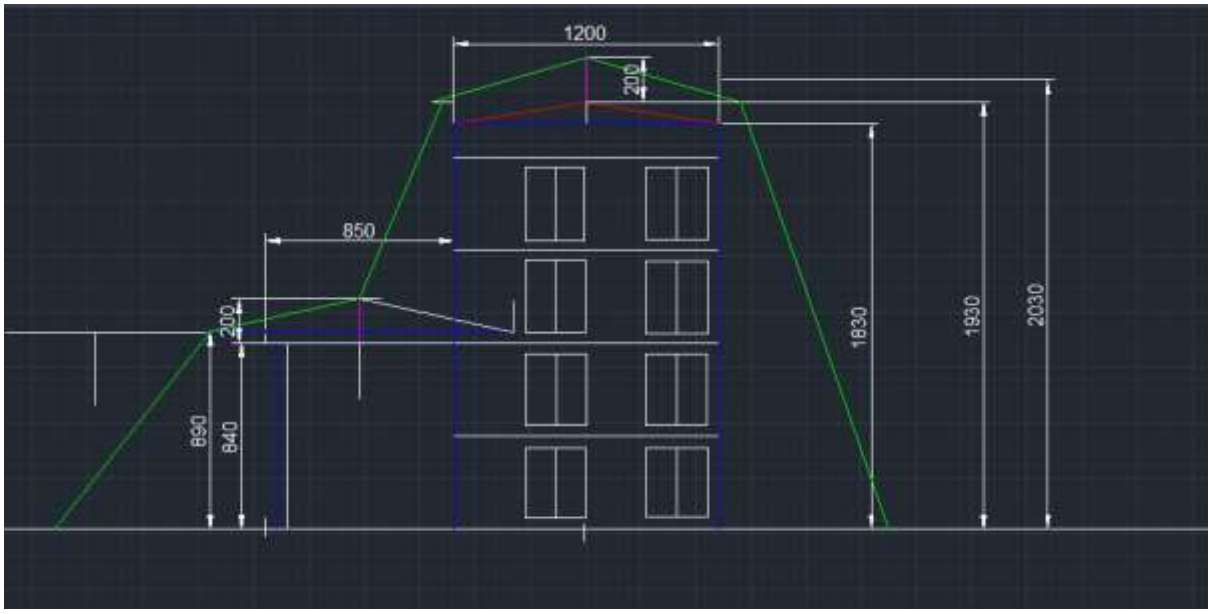


Figure 2.16.6: Side view with termination system wiring

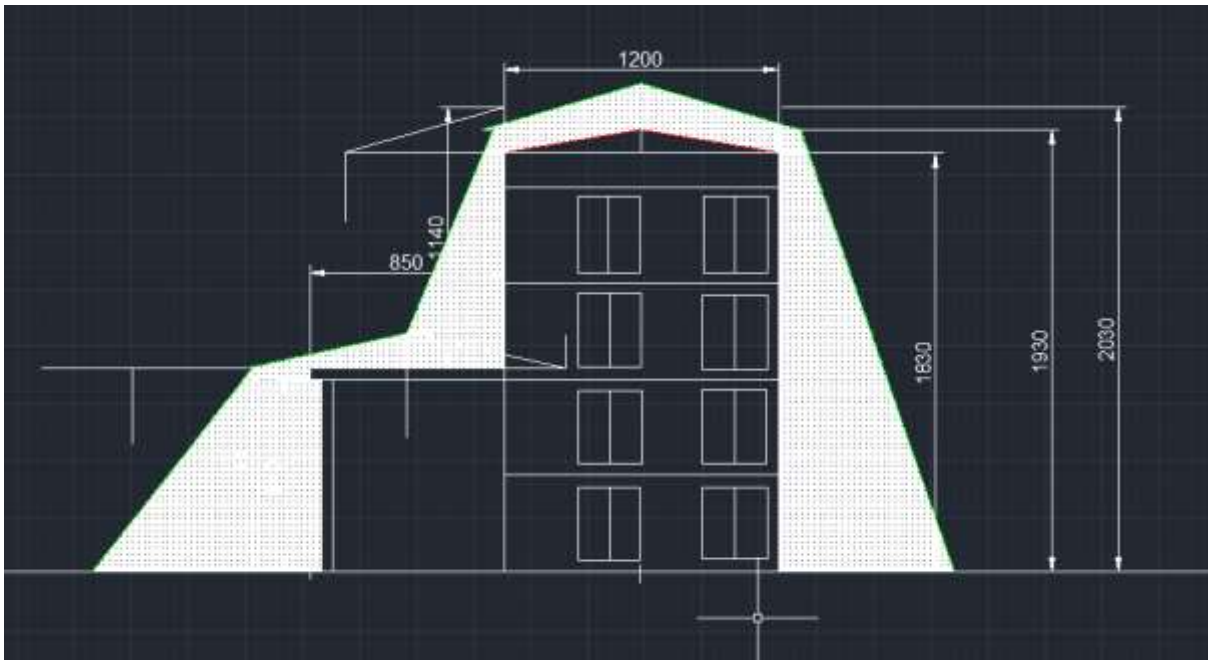


Figure 2.16.7: Side view protected area

C. Top Plan

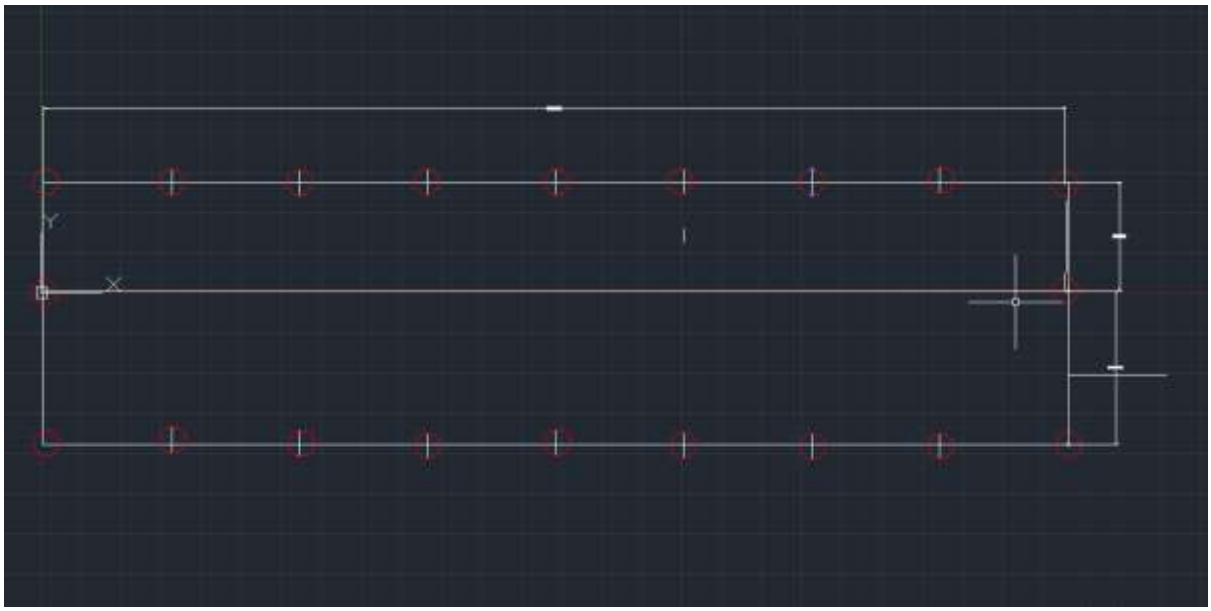


Figure 2.16.8. bottom termination view

The lightning protection system for the Mechanical Workshop was designed using the Protective Angle Method with LPS Class III selected as the design basis. A vertical air-termination rod system was proposed to protect the roof, with the air terminals arranged along the building according to the calculated protection coverage. The lightning current is conducted safely to earth through down conductors distributed around the building, while a Type B closed-loop earth ring electrode is selected as the earth-termination system and

connected to the down conductors. The final earth-electrode dimensions will be determined after obtaining the site soil resistivity and carrying out the required earth-termination testing. Overall, the proposed arrangement provides a systematic path for intercepting, conducting, and safely dissipating lightning current while reducing the risk of physical damage to the workshop.

4.Conclusion

The design task successfully developed a comprehensive electrical solution for the Mechanical Engineering Workshop by considering the electrical installation, solar PV system, and lightning protection system. The electrical installation design addressed the power demand, three-phase balancing, voltage drop, protective devices, incomer rating, and cable selection, while the solar PV design considered the available area, panel and inverter selection, layout, protection, and grid connection. The lightning protection system was designed using the Protective Angle Method, with suitable air-termination, down-conductor, and earth-termination arrangements selected. Overall, the proposed designs provide a safe, reliable, and efficient electrical system for the Mechanical Engineering Workshop.

References

[1] International Electrotechnical Commission, *IEC 60364-5-52: Low-Voltage Electrical Installations—Part 5-52: Selection and Erection of Electrical Equipment—Wiring Systems*, IEC, 2009, amended 2024.

[2] International Electrotechnical Commission, *IEC 60364-4-43: Low-Voltage Electrical Installations—Part 4-43: Protection for Safety—Protection Against Overcurrent*, IEC, 2023.

[3] International Electrotechnical Commission, *IEC 62305-3: Protection Against Lightning—Part 3: Physical Damage to Structures and Life Hazard*, 3rd ed., IEC, 2024.

[4] Sri Lanka Standards Institution, *SLS 1472: Protection Against Lightning*, Parts 1–4, SLSI, Sri Lanka.

[5] International Electrotechnical Commission, *IEC 62548-1: Photovoltaic (PV) Arrays—Part 1: Design Requirements*, IEC, 2023, amended 2025.

[6] International Electrotechnical Commission, *IEC 61730-1: Photovoltaic (PV) Module Safety Qualification—Part 1: Requirements for Construction*, IEC, 2023.

[7] International Electrotechnical Commission, *IEC 62109-1: Safety of Power Converters for Use in Photovoltaic Power Systems—Part 1: General Requirements*, IEC, 2010.

[8] International Electrotechnical Commission, *IEC 61643-11: Low-Voltage Surge Protective Devices—Part 11: Surge Protective Devices Connected to AC Low-Voltage Power Systems—Requirements and Test Methods*, IEC, 2025.

[9] Sri Lanka Sustainable Energy Authority, *Guideline on Rooftop Solar PV Installation in Sri Lanka*, Sri Lanka Sustainable Energy Authority, Sri Lanka.

[10] International Electrotechnical Commission, *IEC 60364-4-41: Low-Voltage Electrical Installations—Part 4-41: Protection for Safety—Protection Against Electric Shock*, IEC.